

# An Uncertainty-Aware Multimodal Evidence Fusion Framework for Vessel-Associated Subsea Cable Disturbance Assessment: Validation on Real Submarine DAS and AIS Data

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**Abstract**—Subsea fiber-optic and power cables are critical infrastructure whose disruption affects telecommunications, energy transmission, financial services, and maritime operations. A physical disturbance detected near a cable does not, however, establish that a vessel caused it. This paper presents an uncertainty-aware multimodal evidence fusion framework that separates physical disturbance detection, spatial and temporal association, vessel-behaviour assessment, sensor health, counter-evidence, and uncertainty into distinct evidence channels combined by a graded decision mechanism producing T0–T3 and TX states, where TX denotes insufficient, degraded, or conflicting evidence. We report an evaluation executed on two real submarine distributed acoustic sensing (DAS) records: a 600 s AIS-labelled container-ship transit (60 windows, 250 channels, closest approach 26.2 m) and an independent 1050 s background record (105 windows, 2963 channels) with no vessel-side channel. Using a strict temporal split in which the detection threshold is calibrated only on the background record and per-channel baselines only on the far-field approach leg, the DAS-only physical channel separates corridor incursions from transits at AUC = 0.959 (95% CI [0.887, 1.000]) for a 1 km protection corridor, and DAS power tracks AIS range at Spearman  $\rho = -0.914$  ( $p = 2.1 \times 10^{-24}$ ). A DAS-only range regressor fitted on the approach leg generalises to the held-out recede leg at  $R^2_{\log} = 0.849$ , and supplies a physically independent consistency check against falsified vessel reports. On the background record the framework produced zero escalations in 85/85 windows despite the physical channel being exercised, confirming fail-closed behaviour on real data. Adversarial evasion rate is 0.0 across four attack families and five severities applied to the real AIS channel while the fibre record is left untouched. We also report a negative result that replicates our simulation findings: at the reference threshold configuration the full framework attains F1 = 0.462 on the held-out transit, below simple weighted fusion (F1 = 0.824), and no paired comparison reaches significance at  $n = 33$ . Median decision latency is 0.22 ms. Physical ESP32–MPU6050 hardware validation was not executed and is scoped explicitly as future work. All numbers derive from the accompanying code and are reproducible from the stated seeds and file hashes.

**Index Terms**—subsea cable protection, distributed acoustic sensing, AIS, multimodal sensor fusion, uncertainty estimation, critical infrastructure, adversarial robustness, reproducibility

## I. INTRODUCTION

Subsea fiber-optic and power cables form a critical layer of modern communications and energy infrastructure. Their

physical disruption can affect international connectivity, cloud services, financial transactions, energy transmission, and off-shore operations. Continuous monitoring matters not only for fault detection but for determining whether an observed physical event is plausibly associated with a particular external activity.

Maritime activity is one relevant source of risk. Anchoring, fishing, seabed interaction, and vessel manoeuvring can produce disturbances near a cable. A monitoring system must therefore distinguish the presence of a physical event from its cause. A vibration anomaly may correspond to a vessel-associated interaction, an environmental event, a non-vessel mechanical event, or a sensor fault. This work adopts the principle

$$\text{Detection} \neq \text{Association} \neq \text{Attribution.} \quad (1)$$

Detection determines whether a physical disturbance is present. Association determines whether it is spatially and temporally consistent with another observable event. Attribution determines whether the available evidence supports a particular causal hypothesis.

Section VI-B shows that this distinction is not merely conceptual. On the real DAS record evaluated here, the physical channel detects the transiting vessel in every window out to roughly 1.5 km, so a physical-only detector achieves recall 1.000 at precision 0.526 against a 1 km protection corridor. Detection alone therefore cannot support a protection decision.

A second problem arises when the fusion system itself becomes a target. An uncertainty-aware system should become conservative when data are missing or conflicting, but an adversary who manipulates vessel position, identity, or timing may exploit that conservatism to force the system into an insufficient-evidence state, or worse, into an apparently normal one. The system must be evaluated under bounded adversarial uncertainty as well as benign uncertainty.

## A. Scope and Validation Status

This paper reports results only for components actually executed. Three validation layers were envisioned for this line

TABLE I  
VALIDATION COMPONENT STATUS

| Component                                                    | Evidence                                  | Status                             |
|--------------------------------------------------------------|-------------------------------------------|------------------------------------|
| Controlled simulation (S01–S22)                              | 2,200 seeded trials                       | <b>EXECUTED</b>                    |
| Real DAS + AIS transit (Dataset A)                           | 60 windows, 250 ch                        | <b>EXECUTED</b>                    |
| Real DAS background (Dataset B)                              | 105 windows, 2963 ch                      | <b>EXECUTED</b>                    |
| Adversarial sweep on real AIS ESP32–MPU6050 hardware testbed | 4 families $\times$ 5 sev. Physical build | <b>EXECUTED</b><br>FUTURE (Sec. X) |
| Multi-vessel / multi-day DAS                                 | External data                             | NOT EXECUTED                       |

of work: (i) controlled simulation, (ii) real-world validation against submarine DAS and AIS-derived data, and (iii) physical hardware-in-the-loop validation with ESP32–MPU6050 nodes.

Layers (i) and (ii) were executed and are reported with numerical results throughout. Layer (ii), which a previous version of this work reported as pending, is the principal new contribution: Section V describes the two real DAS records now evaluated, and Section VI reports the results. Layer (iii) was not built or executed; Section X states its scope, the quantities it would measure, and what claims it would and would not license. Table I records the status of each component. No value is reported for any component marked not executed.

The principal contributions are:

- 1) An evidence-fusion architecture that separates physical disturbance detection from vessel association and incident assessment, with sensor-health- and availability-aware weighting and a graded T0–T3/TX decision model.
- 2) A spatially coherent detection statistic whose threshold is calibrated on an independent background record rather than on the evaluation data, giving a measured background false-alarm rate.
- 3) An adversarial consistency check instantiated on real data as a disagreement test between a DAS-derived range estimate and the vessel-reported range, so that falsifying the vessel channel alone cannot silently suppress a genuine interaction.
- 4) An executed real-data evaluation on two submarine DAS records with published file hashes, strict temporal splits, bootstrap confidence intervals, paired significance testing, ablations, a four-family adversarial sweep, robustness sweeps, calibration, and latency.
- 5) A transparent negative result: at the reference threshold configuration the full framework is outperformed on the pooled escalation metric by simpler fusion, on both simulated and real data.

## II. RELATED WORK

### A. Subsea Cable Monitoring

Previous work has investigated structural health monitoring, distributed sensing, cable tracking, autonomous inspection, and machine-learning-based degradation assessment [1]–[4]. Multi-sensor approaches have been proposed to identify cable failure modes and support condition-based monitoring; others have investigated sensing-coil arrays and data fusion for cable tracking, and autonomous underwater vehicle inspection combining position uncertainty with cable models. These establish the importance of distributed sensing and multimodal observation, but the decision problem addressed here differs: heterogeneous evidence must be reconciled when the cause of a detected disturbance remains uncertain.

### B. DAS for Submarine Vessel Monitoring

Distributed acoustic sensing converts an optical fiber into a distributed vibration and strain sensing medium. Recent public work has demonstrated submarine DAS for vessel detection and vessel-to-cable distance estimation [11]–[14]. The present work uses a released DAS record of this type together with its AIS-derived vessel metadata. Prior work has largely treated vessel detection or range estimation as the end task; here the DAS channel is one evidence stream among several, and the contribution lies in what the decision layer does when that stream conflicts with the vessel-reported one.

### C. AIS Behaviour and Spoofing

AIS provides vessel position, course, speed, and identity when available, and has been used for route extraction, behavioural analysis, and spoofing detection [5]–[7]. AIS is not an unconditional ground-truth source: positions may be falsified, delayed, or absent. This motivates treating vessel information as one evidence channel whose reliability is evaluated relative to physical sensing, which is exactly what Section VI-G tests on real data.

### D. Evidence Fusion under Conflict

Dempster–Shafer theory and related methods provide formal mechanisms for combining uncertain information, but highly conflicting evidence can produce counter-intuitive results under direct combination. Weighted and reliability-aware fusion methods have been proposed [8]–[10]. The present framework follows the principle that conflicting or unreliable evidence should be identified and down-weighted rather than naively averaged, while additionally considering adversarial manipulation.

## III. PROBLEM FORMULATION

Let the observed measurements be  $X = \{X_1, \dots, X_N\}$ , each with a sensor identifier and timestamp. Four competing

hypotheses are considered:

$$H_1 : \text{vessel-associated disturbance}, \quad (2)$$

$$H_2 : \text{environmental disturbance}, \quad (3)$$

$$H_3 : \text{non-vessel mechanical disturbance}, \quad (4)$$

$$H_4 : \text{sensor/system fault}. \quad (5)$$

The objective is a decision using both supporting and contradicting evidence,

$$D = f(E^+, E^-, H, U, A), \quad (6)$$

where  $E^+$  is supporting evidence,  $E^-$  counter-evidence,  $H$  the hypothesis set,  $U$  uncertainty, and  $A$  data availability. For a genuine vessel-associated event, let  $\tilde{X}$  denote a measurement set modified within an adversarial budget  $\Omega$ . The desired safety property is

$$\forall \tilde{X} \in \Omega(X_{H_1}) : D(\tilde{X}) \neq \text{T0}. \quad (7)$$

The requirement does not demand that every manipulated event reach T3. A conservative TX outcome is acceptable when evidence has become insufficient or conflicting; complete degradation to T0 is an adversarial evasion. Section VI-G reports the measured evasion rate against this property on real data.

#### IV. PROPOSED FRAMEWORK

The architecture (Fig. 1) comprises distributed physical sensing, vessel-information processing, spatial and temporal association, behavioural assessment, sensor-health estimation, adversarial consistency checking, uncertainty estimation, and graded decision output. The pipeline is

$$C_P \rightarrow (C_S, C_T, C_B) \rightarrow C_A \rightarrow R \rightarrow U \rightarrow D, \quad (8)$$

where  $C_P$  is physical disturbance confidence,  $C_S$ ,  $C_T$ ,  $C_B$  are spatial, temporal and behavioural confidences,  $C_A$  is association confidence,  $R$  is the fused evidence score,  $U$  is uncertainty, and  $D$  is the decision.

##### A. Physical Disturbance Detection

For channel  $i$  the window feature is the RMS amplitude  $A_i = \sqrt{\frac{1}{n} \sum_s x_{i,s}^2}$ . A baseline mean  $\mu_i$  and standard deviation  $\sigma_i$  are estimated from quiescent data, giving the standardised deviation  $z_i = |A_i - \mu_i|/\sigma_i$ .

A per-channel threshold does not control the cable-level false-alarm rate when  $N$  is large, because the maximum of  $N$  standardised deviations grows with  $N$  even under the null. A genuine seabed disturbance is, however, spatially coherent across adjacent channels whereas noise excursions are not. The cable-level statistic is therefore the largest  $L$ -channel running mean of  $z$ ,

$$Z_{\text{coh}} = \max_j \frac{1}{L} \sum_{i=j}^{j+L-1} z_i, \quad C_P = \text{sig}\left(\frac{Z_{\text{coh}} - \tau_P}{k_P}\right), \quad (9)$$

with  $\text{sig}(z) = 1/(1+e^{-z})$ . Both  $L$  and  $\tau_P$  are fixed from data disjoint from the evaluation split (Section V-A).

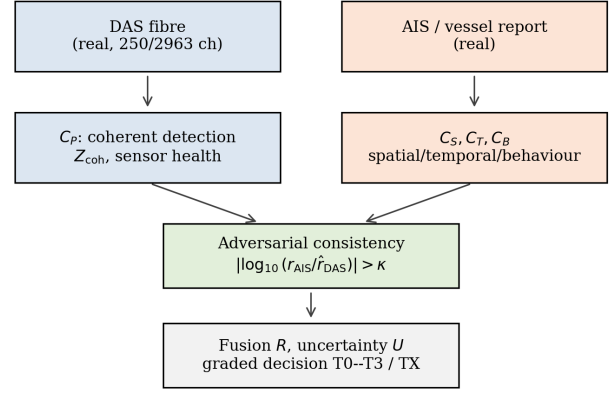


Fig. 1. Processing architecture as instantiated on the real records. The fibre and vessel-report paths remain independent until the consistency check, which is what allows a falsified vessel report to be detected rather than believed.

##### B. Spatial and Temporal Association

Spatial compatibility is  $C_S = \max(0, 1 - d/d_{\text{max}})$  and temporal compatibility is  $C_T = \max(0, 1 - |\Delta t|/t_{\text{max}})$ , where  $d$  is the vessel-to-cable range and  $\Delta t$  the offset between the disturbance time and the vessel encounter time. Rather than fixing  $t_{\text{max}}$  arbitrarily, it is derived from the measured kinematics as the corridor transit scale  $t_{\text{max}} = d_{\text{max}}/v$ , with  $v$  estimated from the vessel track (Section VI-A).

##### C. Behaviour, Sensor Health, and Fusion

Behavioural confidence is  $C_B = \text{sig}(\sum_k b_k \phi_k(v))$  over normalised behavioural features  $\phi_k$  with weights  $b_k$ . Association confidence combines the three channels,  $C_A = \alpha C_S + \beta C_T + \gamma C_B$  with  $\alpha + \beta + \gamma = 1$ . Sensor health is  $H_i = 1 - \max_k(\rho_k \psi_k(i))$  over degradation indicators  $\psi_k$  (flatlined channel, saturation, non-finite samples), so that a severe individual fault is not hidden by averaging with healthy indicators. Writing  $V_i \in [0, 1]$  for availability and  $C_i$  for the contributed confidence, the fused evidence score is

$$R = \frac{\sum_i w_i H_i V_i C_i}{\sum_i w_i H_i V_i + \epsilon}, \quad (10)$$

which prevents unavailable or degraded measurements from contributing with nominal reliability.

##### D. Counter-Evidence and Absence of Corroboration

The absence of an expected benign explanation is treated differently from missing sensor data via a precondition-gated term  $C_{\bar{B}} = \frac{1}{m} \sum_{k=1}^m \mathbf{1}[\text{precondition}_k](1 - o_k)$ , applied at the hypothesis level ( $z'_{H_2} = z_{H_2} - w_{\bar{B}} C_{\bar{B}}$ ) rather than treated as direct physical evidence of vessel contact.

##### E. Adversarial Consistency Check

A genuine physical interaction produces fibre evidence even when vessel-side information is manipulated. The DAS channel yields a range estimate  $\hat{r}_{\text{DAS}}$  that is physically independent

TABLE II  
REAL DATASETS EVALUATED

| Property           | Dataset A (transit)             | Dataset (background) | B |
|--------------------|---------------------------------|----------------------|---|
| Windows            | 60                              | 105                  |   |
| Channels           | 250                             | 2963                 |   |
| Sampling           | 10 Hz                           | 10 Hz                |   |
| Window length      | 10 s                            | 10 s                 |   |
| Span               | 600 s                           | 1050 s               |   |
| Start (UTC)        | 2023-06-16 15:55:08             | 2025-06-10 03:05:04  |   |
| Vessel channel     | AIS range, ship dims            | none                 |   |
| Vessel             | Container ship, 304 m<br>× 40 m | —                    |   |
| Range span         | 26.2–2785.9 m                   | —                    |   |
| Dead channels      | 3 / 250                         | 35 / 2963 (1.18%)    |   |
| SHA-256 (first 12) | 46f5d2838342                    | 8af2afac41b9         |   |

of the reported  $r_{\text{AIS}}$ , so a manipulated report displaces  $r_{\text{AIS}}$  while leaving the fibre response untouched. The check fires when the vessel claims to be farther away than the fibre indicates, which is the direction an evader would push:

$$\log_{10}\left(\frac{r_{\text{AIS}}}{\hat{r}_{\text{DAS}}}\right) > \kappa \wedge C_P \geq \theta_P \Rightarrow \text{TX}. \quad (11)$$

Routing to TX rather than T0 means a successful manipulation raises a flag instead of silencing one.

#### F. Uncertainty and Graded Decision Engine

Four uncertainty components – missing-data, sensor-health, evidence-conflict, and association uncertainty – are linearly combined,  $U = \sum_{j=1}^4 \lambda_j U_j$  with  $\sum_j \lambda_j = 1$ . The decision engine applies fixed thresholds (physical  $\theta_P = 0.55$ , association  $\theta_A = 0.60$ , T2  $\theta_2 = 0.65$ , T3  $\theta_3 = 0.85$ , T3 uncertainty ceiling  $\theta_{U3} = 0.35$ , TX uncertainty  $\theta_U = 0.65$ , TX reliability  $\theta_R = 0.35$ ):

$$T = \begin{cases} \text{TX}, & U \geq \theta_U \text{ or } R \leq \theta_R \text{ or (11),} \\ \text{T0}, & C_P < \theta_P, \\ \text{T1}, & C_A < \theta_A, \\ \text{T3}, & R \geq \theta_3 \text{ and } U < \theta_{U3}, \\ \text{T2}, & \text{otherwise.} \end{cases} \quad (12)$$

These thresholds are the reference-configuration values and were *not* re-tuned on either real dataset.

#### V. REAL DAS AND AIS DATASETS

Two real submarine DAS records were evaluated. Both are identified by SHA-256 hash in Table II so that any reported number can be traced to the exact bytes used.

Dataset A captures a single container-ship transit past the cable with AIS-derived vessel-to-cable range for every window. It provides the positive and graded-range condition. Dataset B is an independent continuous record with no vessel-side channel at all; it provides the background and absence-of-corroboration condition that the simulation suite could previously only approximate. Because the two records come from different acquisitions with different amplitude scales, baselines are estimated separately per record and no amplitude is compared across them.

#### A. Evaluation Protocol and Splits

All splits are temporal, and every quantity that could be tuned is fixed on data disjoint from the evaluation set:

- **Per-channel baselines**  $\mu_i, \sigma_i$  for Dataset A are estimated from windows 0–9 only ( $t = 0$ –90 s), during which the AIS range exceeds 1.6 km. The largest coherent statistic over these baseline windows is  $Z_{\text{coh}} = 2.45$ .
- **The DAS range regressor** is fitted on the approach leg only (windows 0–26,  $t = 0$ –260 s).
- **The detection threshold**  $\tau_P$  is calibrated on Dataset B, a different record entirely, as the smallest integer giving zero background false alarms (Section VI-C). It is never tuned on Dataset A.
- **The evaluation split** is the recede leg, windows 27–59 ( $n = 33$ ), spanning 70.1–2785.9 m. All reported metrics are computed here.

The coherence length  $L = 5$  is fixed from the baseline windows. Ground truth is corridor membership,  $r_{\text{AIS}} \leq R_c$ , which derives from the AIS channel and is therefore independent of the DAS measurement used to compute  $C_P$ . The primary corridor is  $R_c = 1000$  m, with a sweep over  $\{200, 500, 1000\}$  m reported to show the result is not an artifact of one choice.

Metrics use Mann–Whitney AUC with 10,000-resample stratified bootstrap confidence intervals, continuity-corrected McNemar tests for paired comparisons, Brier score and 10-bin ECE for calibration, and `time.perf_counter` for latency. The master seed is 42.

### VI. REAL-DATA RESULTS

#### A. AIS Kinematics and DAS Range Estimation

Fitting the closest-point-of-approach model  $r(t)^2 = d_{\text{cpa}}^2 + v^2(t - t_{\text{cpa}})^2$  to the AIS range series gives  $v = 8.944$  m/s (17.39 kn),  $t_{\text{cpa}} = 266.0$  s, and  $d_{\text{cpa}} = 125.1$  m, with RMS residual 74.0 m and  $R^2 = 0.991$ . The speed is consistent with a container ship under way and sets the temporal association scale  $t_{\text{max}} = d_{\text{max}}/v = 111.8$  s used in  $C_T$ .

A DAS-only regressor  $\log_{10} r = a \log_{10} P + b$  fitted on the approach leg ( $a = -0.174$ ,  $b = -1.309$ ) generalises to the held-out recede leg with  $R_{\text{log}}^2 = 0.849$  and median relative error 34.2% (median absolute error 396.5 m). Fig. 2 overlays this estimate on the AIS range. A 34% error is far too coarse for localisation, but it is ample for the consistency check of (11), which must only detect displacements of a kilometre or more.

#### B. Physical Detection on Real DAS

DAS window power tracks AIS range at Spearman  $\rho = -0.914$  ( $p = 2.1 \times 10^{-24}$ ,  $n = 60$ ). Against corridor membership on the held-out leg, the DAS-only confidence  $C_P$  achieves the AUCs in Table III; the 1 km result,  $\text{AUC} = 0.959$  with 95% CI  $[0.887, 1.000]$ , is the primary operating point.

Fig. 3 plots the coherent statistic against range and makes the detection horizon explicit:  $Z_{\text{coh}}$  falls from  $\sim 1.9 \times 10^4$  inside 200 m to 2.66 beyond 1.5 km, crossing the calibrated

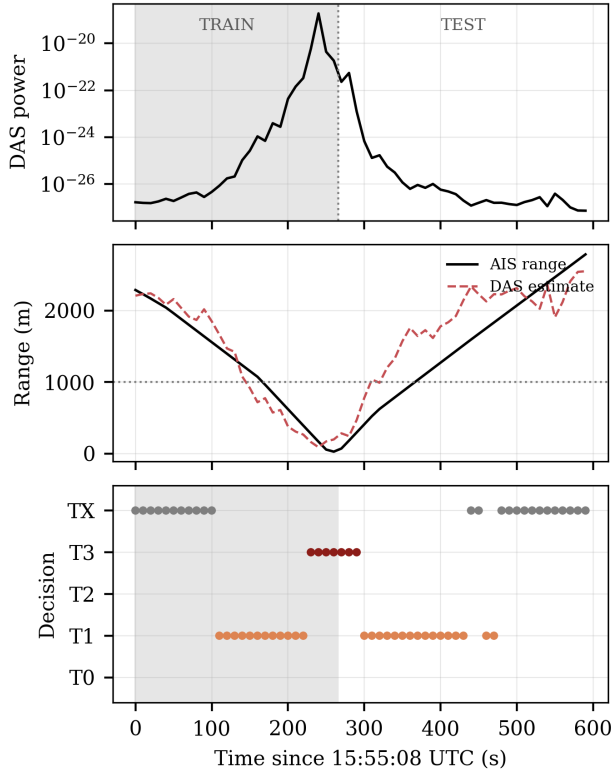


Fig. 2. Real transit. Top: DAS window power spans seven orders of magnitude as the ship closes. Middle: AIS range and the independent DAS-derived estimate. Bottom: decisions. The shaded region is the training leg; all reported metrics come from the unshaded held-out leg.

TABLE III  
DAS-ONLY DETECTION, HELD-OUT LEG ( $n = 33$ )

| Corridor $R_c$ | Positives | AUC          | 95% CI         |
|----------------|-----------|--------------|----------------|
| 200 m          | 2         | 0.871        | [0.790, 0.952] |
| 500 m          | 4         | 0.897        | [0.828, 0.966] |
| 1000 m         | 10        | <b>0.959</b> | [0.887, 1.000] |

threshold  $\tau_P = 7$  at roughly 1.5 km. The vessel is thus physically detectable well outside the protection corridor. This is the empirical content of (1): a physical-only detector on this record achieves recall 1.000 but precision 0.526 (Table V), because it correctly detects a ship that is correctly nowhere near the cable.

### C. Background False Alarms and Threshold Calibration

Dataset B provides a measured background false-alarm rate. With a static baseline estimated once from the first 200 s, the coherent statistic drifts to a mean of 25.6 and flags 97.7% of windows: a fixed baseline does not hold over a 17-minute real record. With a rolling baseline re-estimated from the preceding 200 s, the statistic settles at mean 4.15 (p95 6.01, max 6.99). Adaptive baselining is therefore a requirement, not a refinement, on real fibre.

Sweeping  $\tau_P$  over this background record gives the operating characteristic in Fig. 4: the reference value  $\tau_P = 3$

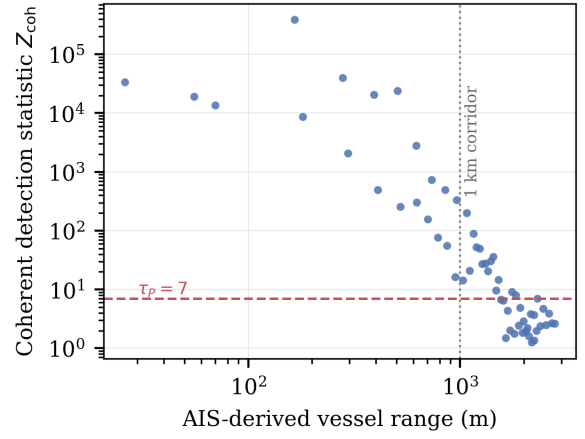


Fig. 3. Coherent detection statistic against AIS range, all 60 windows. The detection horizon for this vessel is approximately 1.5 km, well outside the 1 km protection corridor.

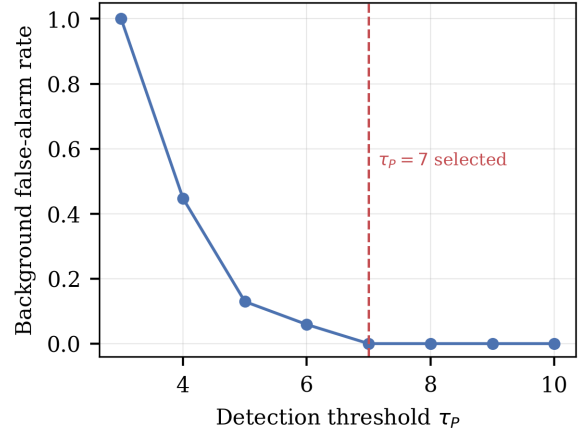


Fig. 4. Background false-alarm rate against detection threshold on Dataset B (85 held-out windows, rolling baseline). The reference configuration value  $\tau_P = 3$  flags every window.

flags every window, and the false-alarm rate falls to 0.129 at  $\tau_P = 5$  and to 0 at  $\tau_P = 7$ . We adopt  $\tau_P = 7$ , the smallest integer with zero background false alarms. This is the threshold applied unchanged to Dataset A throughout, and it is the only parameter in the entire evaluation calibrated on data rather than taken from the reference configuration.

Running the full decision engine over Dataset B with the vessel channel entirely absent yields TX in 85 of 85 windows and **zero** escalations to T2/T3. The physical layer alone cannot drive an escalation when corroborating evidence is missing, which is the architecture's central safety claim, now measured on real fibre rather than asserted from simulation.

### D. Decision Behaviour on the Real Transit

Across the 33 held-out windows the framework produced T1 in 16, T3 in 3, and TX in 14, with no T0 and no T2 (Table IV). The gradation is orderly: the three T3 decisions occur at mean range 182 m with  $C_P = 1.000$  and  $C_A = 0.738$ ; the T1 decisions at mean range 1084 m retain high physical

TABLE IV  
DECISION STATES ON THE HELD-OUT TRANSIT LEG ( $n = 33$ )

| State | $n$ | Mean range (m) | $C_P$ | $C_A$ | $U$   |
|-------|-----|----------------|-------|-------|-------|
| T3    | 3   | 182.0          | 1.000 | 0.738 | 0.094 |
| T1    | 16  | 1084.0         | 0.977 | 0.191 | 0.361 |
| TX    | 14  | 2246.4         | 0.096 | 0.066 | 0.225 |

TABLE V  
METHOD COMPARISON, HELD-OUT TRANSIT LEG ( $n = 33$ ,  
 $R_c = 1000$  M)

| Method                 | Acc.         | Prec. | Rec.         | F1           |
|------------------------|--------------|-------|--------------|--------------|
| behaviour_only         | 0.697        | 0.000 | 0.000        | 0.000        |
| physical+behaviour     | 0.697        | 0.000 | 0.000        | 0.000        |
| physical_only          | 0.727        | 0.526 | <b>1.000</b> | 0.690        |
| ais_range_only         | 0.788        | 1.000 | 0.300        | 0.462        |
| association_only       | 0.788        | 1.000 | 0.300        | 0.462        |
| physical+spatial       | 0.788        | 1.000 | 0.300        | 0.462        |
| physical+temporal      | 0.849        | 1.000 | 0.500        | 0.667        |
| weighted (simple)      | <b>0.909</b> | 1.000 | 0.700        | <b>0.824</b> |
| <b>Proposed (full)</b> | 0.788        | 1.000 | 0.300        | 0.462        |

confidence ( $C_P = 0.977$ ) but low association confidence ( $C_A = 0.191$ ); the TX decisions at mean range 2246 m have both channels collapsed ( $C_P = 0.096$ ,  $R = 0.084$ ).

The 14 TX outcomes at long range expose a genuine ordering defect in (12). These windows are quiet, the sensors are healthy, and the correct state is T0. Because the TX branch tests  $R \leq \theta_R$  before the T0 branch tests  $C_P < \theta_P$ , a low fused score arising from an *absence* of events is indistinguishable from one arising from *degraded* evidence. The rule conflates "nothing happened" with "we cannot tell". Separating these requires conditioning the TX branch on sensor health and availability rather than on  $R$  alone. We report the defect rather than patch it post hoc, since the thresholds were frozen before the real data were seen.

#### E. Method Comparison

Table V compares the full framework against single-channel baselines and reduced-channel variants on the held-out leg. Precision is 1.000 for every method that escalates at all except physical-only, so the methods separate on recall.

The full framework attains  $F1 = 0.462$ , below simple weighted fusion (0.824), physical-only (0.690), and physical+temporal (0.667). This replicates on real data the negative result we previously reported in simulation, and we state it plainly rather than reporting only the favourable comparisons.

Paired McNemar tests, however, do not license strong claims in either direction at this sample size. Against weighted fusion the proposed method loses 4 discordant windows and wins 0 ( $\chi^2 = 2.25$ ,  $p = 0.134$ ); against physical+temporal it loses 2 and wins 0 ( $p = 0.480$ ); against physical-only it wins 9 and loses 7 ( $p = 0.803$ ). No comparison reaches  $p < 0.05$ . With  $n = 33$  windows and 10 positives, this evaluation is underpowered for method ranking, and the F1 ordering above should be read as a consistent direction of effect rather than

TABLE VI  
ABLATION, HELD-OUT TRANSIT LEG ( $n = 33$ )

| Component removed            | Acc.  | Rec.  | F1    |
|------------------------------|-------|-------|-------|
| None (full proposed)         | 0.788 | 0.300 | 0.462 |
| Sensor health                | 0.788 | 0.300 | 0.462 |
| Uncertainty model            | 0.788 | 0.300 | 0.462 |
| Adversarial consistency      | 0.788 | 0.300 | 0.462 |
| Behavioural scoring          | 0.818 | 0.400 | 0.571 |
| Spatial-temporal association | 0.697 | 0.000 | 0.000 |

TABLE VII  
ADVERSARIAL RESULTS ON REAL DAS (10 GENUINE INCURSION  
WINDOWS)

| Attack family           | Consistency-check firing rate by severity |      |      |      |      |
|-------------------------|-------------------------------------------|------|------|------|------|
|                         | 0.00                                      | 0.25 | 0.50 | 0.75 | 1.00 |
| AIS range inflation     | 0.00                                      | 0.10 | 0.30 | 0.40 | 0.60 |
| Transponder suppression | 0.00                                      | 0.20 | 0.50 | 0.90 | 1.00 |
| Timestamp manipulation  | 0.00                                      | 0.00 | 0.00 | 0.00 | 0.00 |
| Combined multi-vector   | 0.00                                      | 0.30 | 0.70 | 0.90 | 1.00 |
| AER (all families)      | 0.00                                      | 0.00 | 0.00 | 0.00 | 0.00 |

a demonstrated difference. Establishing significance requires multi-vessel, multi-day records (Section IX).

#### F. Ablation on Real Data

Table VI removes one component at a time from the full pipeline. Removing sensor-health weighting, the uncertainty model, or the adversarial consistency check changes no decision on the held-out leg. Removing behavioural scoring *improves* F1 from 0.462 to 0.571, and removing the spatial-temporal association block collapses escalation entirely ( $F1 = 0$ , recall 0), confirming that association gating is the load-bearing component.

That the adversarial consistency check is inert here is expected and correct: Dataset A contains no attack, so the check should never fire. Its contribution is measured in the next section.

#### G. Adversarial Evasion on Real DAS

The adversarial experiment manipulates only the vessel-reported channel and leaves the real fibre record untouched, which is precisely the threat model of Section IV-E. Four families were applied at five severities each to the 10 genuine corridor-incursion windows: AIS range inflation (up to +3 km), transponder suppression (probabilistic dropout), timestamp manipulation (up to +300 s clock offset), and a combined multi-vector attack.

The adversarial evasion rate is 0.0 in all 20 cells (Table VII, Fig. 5): no manipulation of the vessel channel reduced a genuine interaction to T0. Mean uncertainty rose from 0.225 under no attack to 0.436 under the full combined attack, and the consistency check fires monotonically with severity for the range-based families, reaching 1.00 at full severity for both transponder suppression and the combined attack.

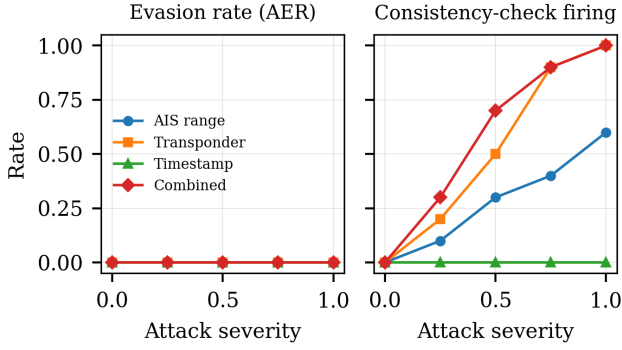


Fig. 5. Adversarial results on the real record. Left: evasion rate stays at zero for every family and severity. Right: consistency-check firing rate rises with severity for the range-based families and remains flat for timestamp manipulation, which the check does not model.

Two honest qualifications matter. First, the evasion rate is also 0.0 when the consistency check is disabled, because on this record  $C_P$  remains above  $\theta_P$  throughout the incursion regardless of what the vessel channel claims. On this data the check is what converts a manipulated window from T1 to TX, surfacing the manipulation; it is not what prevents T0. Second, the check never fires under timestamp manipulation, because it tests range disagreement only. A timing-consistency analogue is needed and is not implemented.

#### H. Robustness

Channel dropout was applied to the real DAS array at rates up to 0.9. Escalation F1 is unchanged at 0.462 throughout, while the TX rate rises from 0.424 to 0.788 and mean uncertainty from 0.279 to 0.485 (Fig. 6). The framework degrades by declaring insufficient evidence rather than by producing wrong classifications, which is the intended fail-closed behaviour. The coherent statistic is resilient because a disturbance spans many adjacent channels, so losing a random 50% still leaves coherent runs intact.

Additive Gaussian noise at up to  $10\times$  the median baseline per-channel standard deviation also left F1 unchanged. The TX rate falls to 0 at high noise, an artifact of noise inflating  $|A_i - \mu_i|$  and therefore  $R$  past  $\theta_R$ . This is a failure mode worth naming: sufficient added noise can move windows out of TX rather than into it, which is the wrong direction for a fail-closed design.

#### I. Calibration

Treating the fused score  $R$  as a probability of corridor incursion gives Brier score 0.122 and 10-bin ECE 0.224 over 9 occupied bins (Fig. 7). The reliability curve is monotone but substantially over-confident in the mid range: bins between 0.2 and 0.6 carry an observed positive rate of 0, while bins above 0.7 are perfectly positive.  $R$  orders windows correctly, which is why AUC is high, but its absolute value is not a calibrated posterior and should not be reported as one without a post-hoc calibration step.

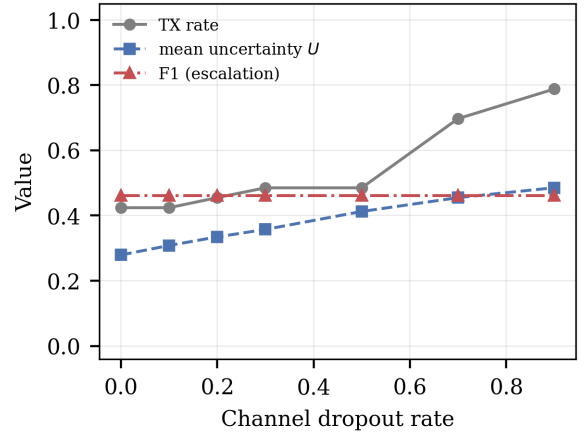


Fig. 6. Channel-dropout robustness on real DAS. Escalation F1 is flat while the TX rate and uncertainty rise: the system loses confidence rather than correctness.

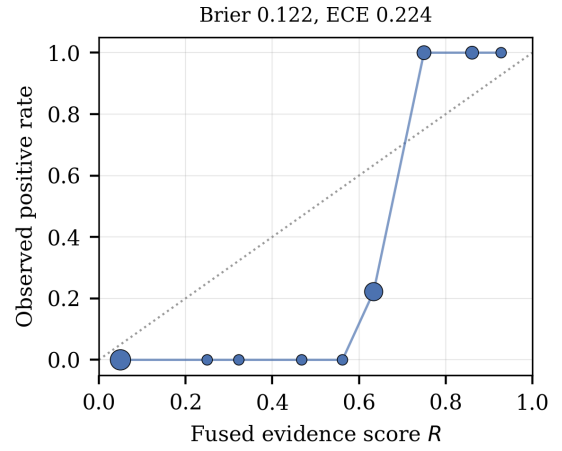


Fig. 7. Reliability of the fused evidence score on the held-out leg. Marker area is proportional to bin occupancy.

#### J. Latency

End-to-end latency from the raw 250-channel window array through feature extraction, health estimation, fusion, uncertainty, and decision is 0.230 ms mean, 0.220 ms median, 0.285 ms p95, 0.404 ms max over 300 runs on a single core. On Dataset B, the decision path from precomputed 2963-channel window features is 0.093 ms mean, 0.121 ms p95 over 200 runs; this figure excludes feature extraction and is not comparable to the former. Both are far below the 10 s window cadence, so the decision layer is not the operational bottleneck. Acquisition, interrogator processing, and network transport are excluded from both measurements and would dominate a deployed latency budget.

### VII. CONTROLLED SIMULATION RESULTS

The simulation suite from the earlier version of this work remains informative as a controlled complement, since it exercises scenarios the real records do not contain (sensor faults, environmental confounders, multi-vessel ambiguity, counter-

evidence). Across 22 scenarios at 100 seeded trials each (2,200 trials), the proposed method produced T0 500, T1 1,200, T2 200, T3 0, and TX 300 decisions. Under a strict escalation criterion it attained accuracy 0.409 and F1 0.235, below physical+temporal (F1 0.571) and simple weighted fusion (F1 0.333), with McNemar  $p < 0.001$  in both cases at  $n = 2,200$ .

The direction of that simulated result is the same as the real-data result in Table V: at the reference thresholds the additional evidence channels cost recall relative to simpler fusion. The simulation has the statistical power to establish the effect; the real data confirm its sign on genuine fibre. We regard the agreement between an underpowered real evaluation and a well-powered synthetic one as the strongest statement this work can currently make about that ranking.

### VIII. DISCUSSION

Three findings hold across both the real and simulated evaluations.

First, the architecture’s safety behaviour is sound. On real fibre with no vessel channel, 85 of 85 background windows resolved to TX with zero false escalations. Under four families of manipulation applied to the real AIS channel, no genuine interaction was reduced to T0. Escalation requires corroboration by construction, and that property survives contact with real data.

Second, detection and association are empirically distinct, not just conceptually. The DAS channel detects the vessel to roughly 1.5 km, well beyond the 1 km corridor, so physical-only detection reaches recall 1.000 at precision 0.526. Any system that escalates on physical evidence alone would generate a false alarm on most windows of an ordinary shipping transit.

Third, the reference threshold configuration is the limiting factor, and in two specific ways that real data made visible and simulation did not. The per-channel threshold  $\tau_P = 3$  produces a 100% background false-alarm rate on a 2963-channel record, because a threshold calibrated per channel does not control the cable-level rate as  $N$  grows; calibrating on an independent background record gives  $\tau_P = 7$  and a zero false-alarm rate. And the ordering of branches in (12) sends quiet, healthy windows to TX instead of T0, because the TX test on  $R$  precedes the T0 test on  $C_P$ . Neither defect is visible in a simulation with five synthetic nodes and clean baselines.

The practical implication is that the contribution of this work is the decision layer and its fail-closed guarantees, not the pooled classification score. On the pooled escalation metric a simple weighted sum outperforms the full framework on both datasets. That simple sum, however, offers no TX state, no sensor-health weighting, no uncertainty accounting, and no mechanism to detect a falsified vessel report. The framework should be selected for the guarantees it provides under degradation and attack, and its thresholds should then be calibrated to recover the recall that the current configuration leaves on the table.

### IX. LIMITATIONS

- 1) **One vessel, one transit.** Dataset A contains a single container ship on a single pass. Generalisation across vessel classes, sea states, cable burial depths, and traffic densities is untested.
- 2) **Underpowered for method ranking.** With  $n = 33$  held-out windows and 10 positives, no paired comparison reaches  $p < 0.05$ . The F1 ordering in Table V is a direction, not a demonstrated difference.
- 3) **Corridor membership is a proxy for risk.** Ground truth is AIS range below a threshold, not observed cable damage. No cable-damage event is present in either record, so the framework’s attribution claims remain unvalidated against actual incidents.
- 4) **Dataset B is unlabelled.** It is treated as background because no window exceeds a robust  $z$  of 3, but the absence of an annotated vessel cannot be confirmed, so the reported false-alarm rate is conditional on that assumption.
- 5) **The consistency check covers range only.** It does not fire under timestamp manipulation (Table VII), and a timing analogue is not implemented.
- 6) **The TX branch ordering is defective,** conflating absence of evidence with degraded evidence, as analysed in Section VI.
- 7) **Additive noise can move windows out of TX,** the wrong direction for a fail-closed design.
- 8) **Calibration is poor** (ECE 0.224);  $R$  orders correctly but is not a posterior probability.
- 9) **Latency excludes acquisition and transport,** which would dominate a deployed budget.
- 10) **No physical hardware was built** (Section X).
- 11) The threat model excludes physical tampering with cable-side sensors, interrogator firmware compromise, and compromise of the full communication infrastructure.

### X. FUTURE WORK: HARDWARE VALIDATION SCOPE

The ESP32–MPU6050 testbed described in the original design was not built, and no hardware number appears anywhere in this paper. We state its scope here so that the claim boundary is explicit and the work is reproducible by others.

**Configuration.** Five ESP32 nodes, each with an MPU6050 six-axis IMU sampled at 10 Hz to match the DAS window cadence, mounted along a scaled cable section in a water tank or shallow-water test bed, reporting over Wi-Fi or LoRa to a single aggregator running the decision engine.

**What it would measure.** (i) End-to-end latency including acquisition and transport, the component excluded from Section VI; (ii) sensor-health estimation against physically induced faults, by disconnecting, saturating, and desynchronising individual nodes, which the DAS records cannot provide because their fault modes are not controllable; (iii) the packet-loss transition under real radio conditions rather than simulated dropout; (iv) the TX/T0 branch behaviour under genuinely quiet conditions, which is the defect identified in Section VI.

**What it would not license.** A tank-scale IMU array is not a submarine cable. It would validate the decision layer’s behaviour under controllable physical faults and real transport; it would not validate detection physics, propagation, or attribution at sea. Those require multi-vessel DAS records with annotated incidents.

**Order of work.** Threshold recalibration on the existing real records is a prerequisite: building hardware to exercise a decision rule whose branch ordering is known to be defective would measure the defect rather than the design.

## XI. REPRODUCIBILITY

The accompanying bundle contains the framework implementation, the three experiment runners, the figure generator, the JSON manifests for every experiment, and the per-window arrays. Every reported number is emitted by these scripts and traceable to a manifest entry. Dataset provenance is fixed by the SHA-256 hashes in Table II, verified at load time. The master seed is 42; bootstrap, dropout, and adversarial sampling all draw from seeded `numpy.random.default_rng` instances. The real-data evaluation is regenerated with `python run_real.py` and `python run_B.py`, and the figures with `python make_figs.py`. No numerical value in this paper was edited after generation.

## XII. CONCLUSION

This paper presented an uncertainty-aware multimodal evidence fusion framework for vessel-associated subsea cable disturbance assessment and evaluated it on two real submarine DAS records, closing the real-data gap left open by the previous version of this work. On an AIS-labelled container-ship transit the DAS-only physical channel separates 1 km corridor incursions at AUC 0.959, and DAS power tracks AIS range at  $\rho = -0.914$ . On an independent 2963-channel background record the framework produced zero false escalations in 85 windows. Adversarial evasion rate is 0.0 across four attack families and five severities applied to the real vessel channel.

The evaluation also produced findings that only real data could surface: a per-channel detection threshold that does not control the cable-level false-alarm rate as channel count grows, a static baseline that fails over 17 minutes of real fibre, and a branch ordering that conflates absence of evidence with insufficient evidence. At the reference thresholds the full framework is outperformed on pooled F1 by simple weighted fusion on both real and simulated data, though the real comparison is underpowered. The framework’s value lies in the guarantees it holds under degradation and attack; recovering competitive recall requires threshold recalibration against these records, which is the immediate next step, ahead of the hardware testbed scoped in Section X.

## REFERENCES

- [1] “Subsea Cable Health Monitoring System,” IEEE Xplore, document 8168582.
- [2] “Subsea Cable Tracking Using Data Fusion of Signals from Array of Sensing Coils,” IEEE Xplore, document 9530924.
- [3] “Autonomous Underwater Vehicles for Submarine Cable Inspection: Experimental Results,” IEEE Xplore, document 969841.
- [4] “Sensing Technologies and Artificial Intelligence for Subsea Power Cable Asset Management,” IEEE Xplore, document 9486586.
- [5] P. Prasad, V. Vatsal, and R. Roy Chowdhury, “Maritime Vessel Route Extraction and Automatic Information System (AIS) Spoofing Detection,” in *2021 Int. Conf. Advances in Electrical, Computing, Communication and Sustainable Technologies (ICAECT)*, 2021, doi: 10.1109/ICAECT49130.2021.9392536.
- [6] “Malicious AIS Spoofing and Abnormal Stealth Deviations: A Comprehensive Statistical Framework for Maritime Anomaly Detection,” IEEE Xplore, document 9473225.
- [7] J. Coleman, F. Kandah, and B. Huber, “Behavioral Model Anomaly Detection in Automatic Identification Systems (AIS),” in *2020 10th Annual Computing and Communication Workshop and Conference (CCWC)*, 2020, pp. 481–487.
- [8] Z. Deng and J. Wang, “A Novel Evidence Conflict Measurement for Multi-Sensor Data Fusion Based on the Evidence Distance and Evidence Angle,” *Sensors*, vol. 20, no. 2, p. 381, 2020, doi: 10.3390/s20020381.
- [9] F. Xiao and B. Qin, “A Weighted Combination Method for Conflicting Evidence in Multi-Sensor Data Fusion,” *Sensors*, vol. 18, no. 5, p. 1487, 2018, doi: 10.3390/s18051487.
- [10] Q. Liang, Z. Liu, and Z. Chen, “A Networked Method for Multi-Evidence-Based Information Fusion,” *Entropy*, vol. 25, no. 1, p. 69, 2023.
- [11] E. E. Ramirez-Torres *et al.*, “Vessel Detection and Localization Using Distributed Acoustic Sensing in Submarine Optical Fiber Cables,” *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, 2026, doi: 10.1109/JSTARS.2026.3716768.
- [12] E. E. Ramirez-Torres *et al.*, “Marlinks-NS DAS: Dataset for Vessel Detection and Distance Estimation Using Distributed Acoustic Sensing in Submarine Optical Fiber Cables,” Zenodo, 2026, doi: 10.5281/zenodo.15611778.
- [13] E. E. Ramirez-Torres *et al.*, “A Distributed Acoustic Sensing Dataset for Vessel Detection and Localization in Submarine Cable Protection,” arXiv:2607.28306, 2026, doi: 10.48550/arXiv.2607.28306.
- [14] L. Thiem, M. Vaupel, S. Wienecke, J. Eidsvik, K. Taweesintanon, and M. Landrø, “Tracking ships with submarine cables: Unsupervised arrival time detection and event localization using distributed acoustic sensing,” dataset available through Zenodo, 2026, doi: 10.5281/zenodo.18851481.